

BASIC PRINCIPLES OF MEDICINE 1: **FOUNDATIONS TO MEDICINE**

FTM LECTURE 46

Structural Chromosomal Abnormalities

Presented by: Dr. Shellon Thomas
(sthoma10@sgu.edu)

Dr. Mary Maj
Email: mmaj@sgu.edu

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Pre-Lecture - Required Reading and Objectives

Human Genetics and Genomics, Korf and Irons, 4th Edition Chapter 6

- You can find the following pages and chapters on your Sakia site under:**
- Resources/General Information/Book Access

Lecture Objectives

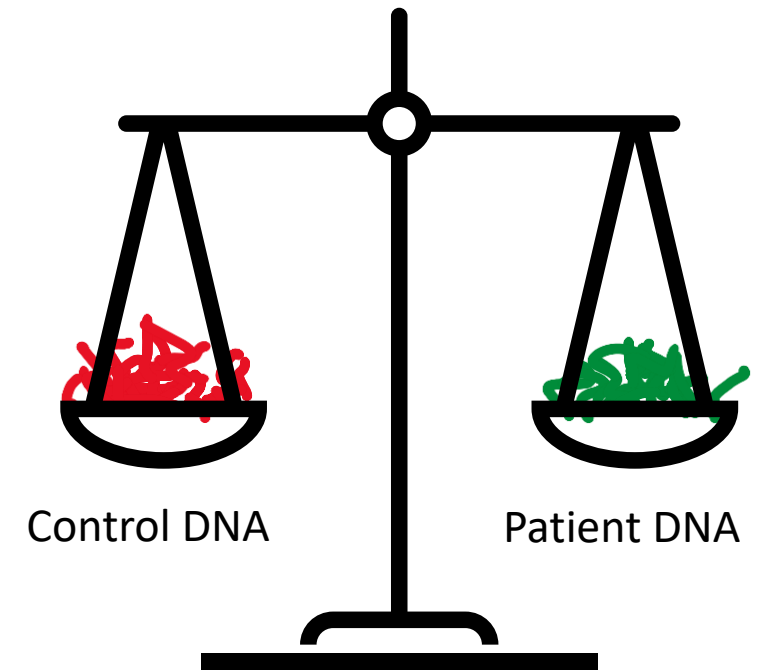
SOM.MK.III.BPM1.1.FTM.3.GNET.0124	Analyze balanced, unbalanced, reciprocal and Robertsonian translocations, describe how a reciprocal translocation can create a fusion gene
SOM.MK.III.BPM1.1.FTM.3.GNET.0125	Describe how the Robertsonian translocation can explain Down syndrome. Indicate the risk that a carrier of a Robertsonian translocation has for having a child with Down syndrome
SOM.MK.III.BPM1.1.FTM.3.GNET.0126	Analyze microdeletion syndromes with the following examples: cri-du-chat, 22q11.2 DS, Wolf_Hirschhorn, Prader-Willi, Angelman
SOM.MK.III.BPM1.1.FTM.3.GNET.0127	Compare and contrast other chromosomal abnormalities including inversions, ring chromosomes, uniparental disomy and isochromosomes
SOM.MK.III.BPM1.1.FTM.3.GNET.0128	Analyze cytogenomic tests used for diagnosis of chromosomal aberrations: Karyotype, FISH, Microarray CGH

Structural Chromosomal Abnormalities

- **Balanced** (No gain or loss of genetic material)
- **Unbalanced** (Gain or loss of genetic material)

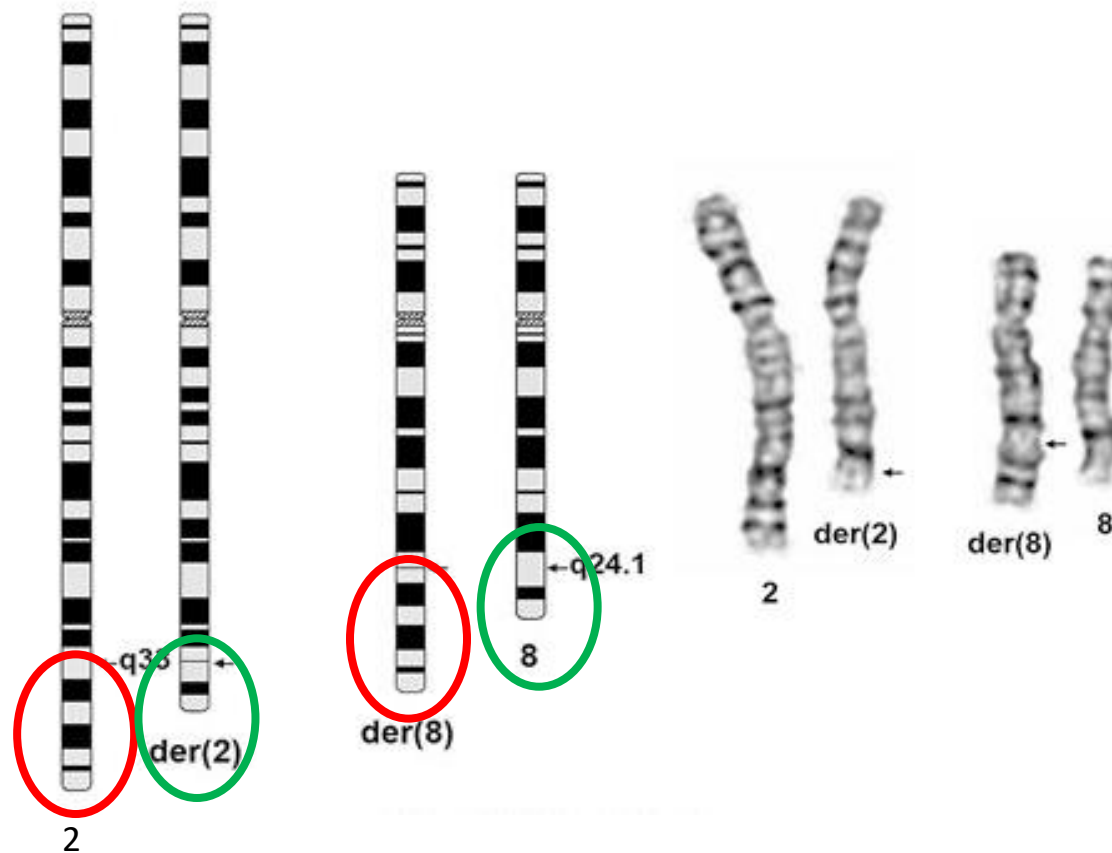
Types of Abnormalities Discussed:

- Translocations
 - Reciprocal
 - Robertsonian
- Deletions
- Inversions
- Ring chromosome
- Isochromosome

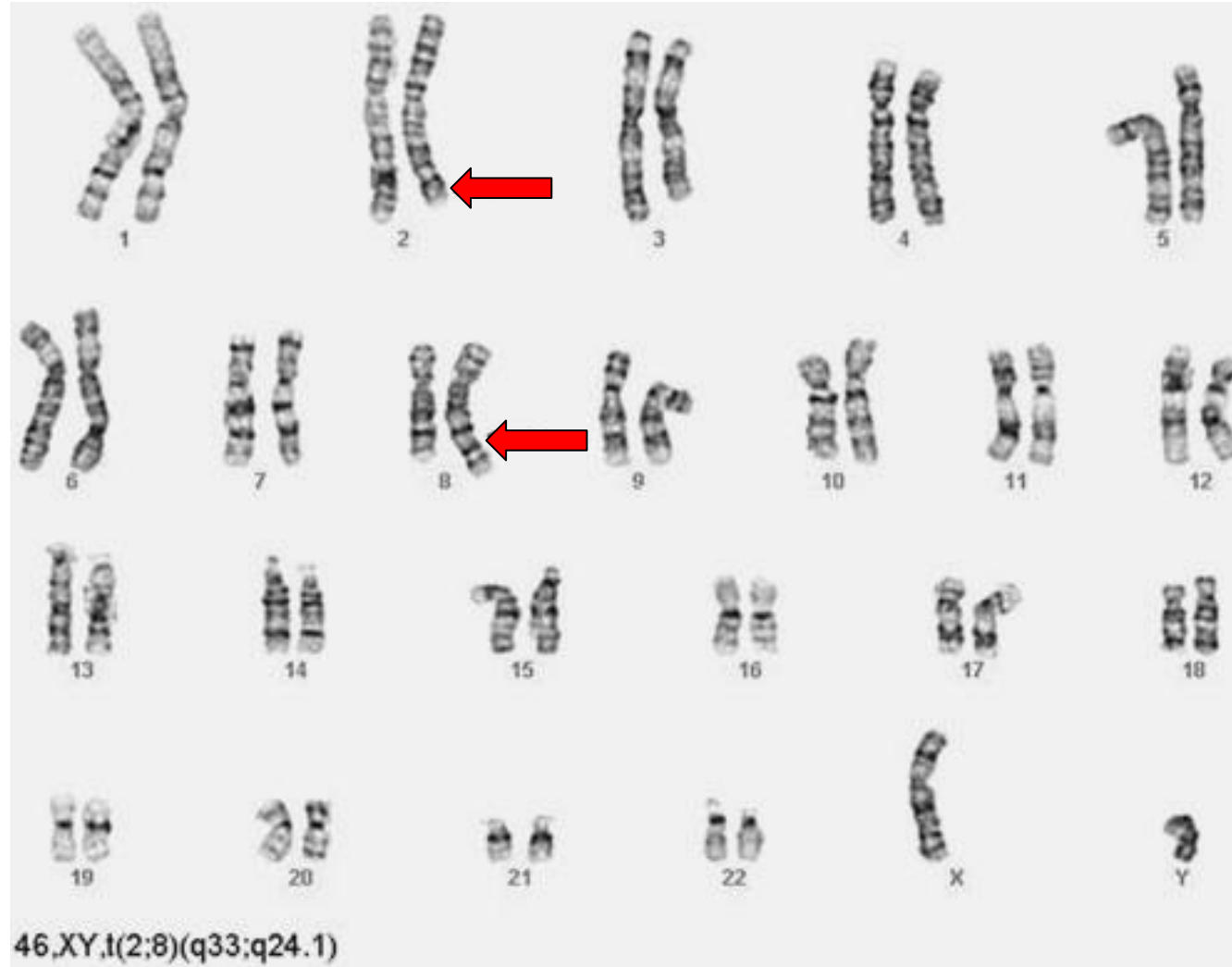


Reciprocal Translocation

- Exchange of genetic material between non-homologous chromosomes



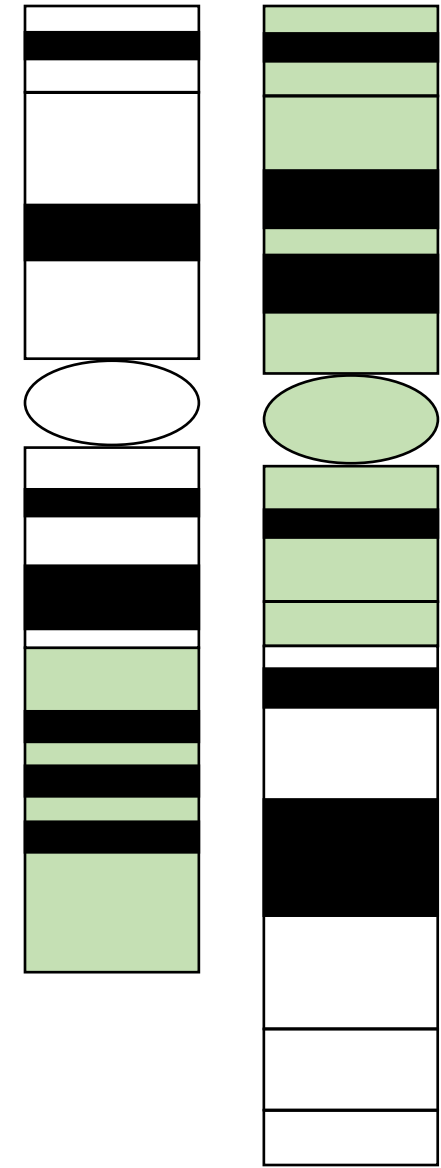
Balanced reciprocal translocation (chromosome 2 & 8)



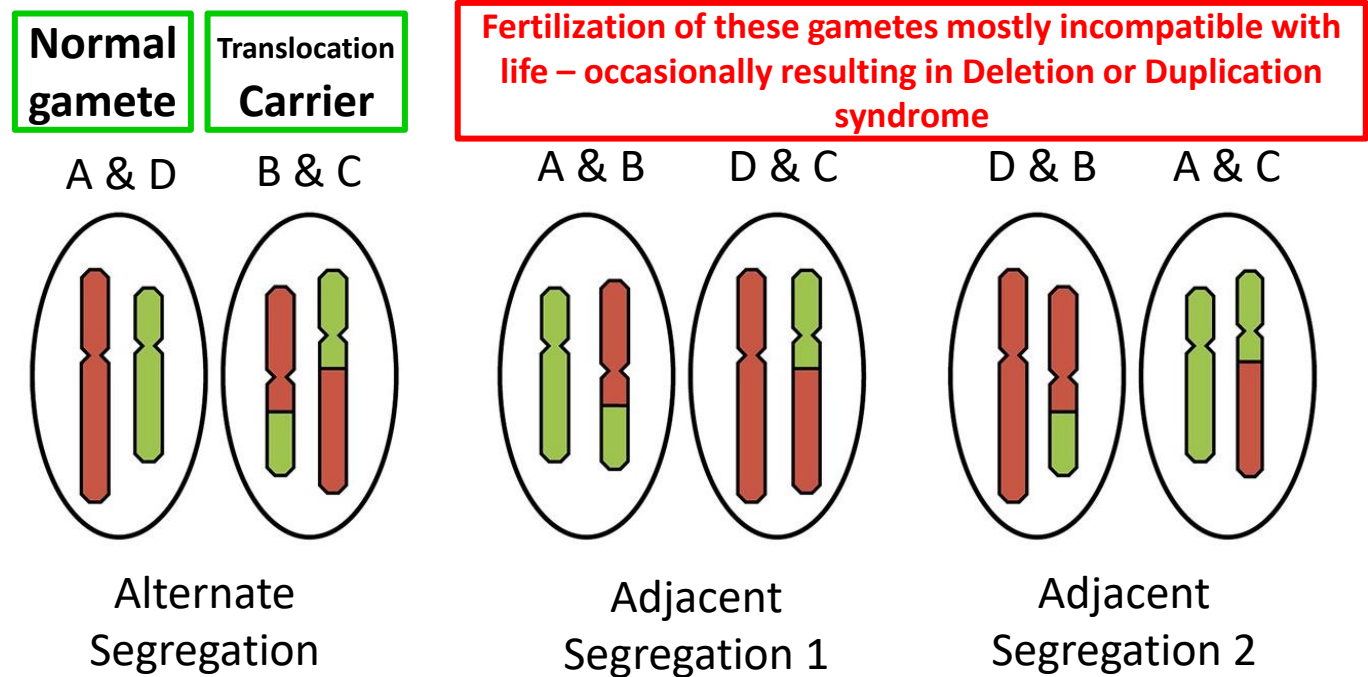
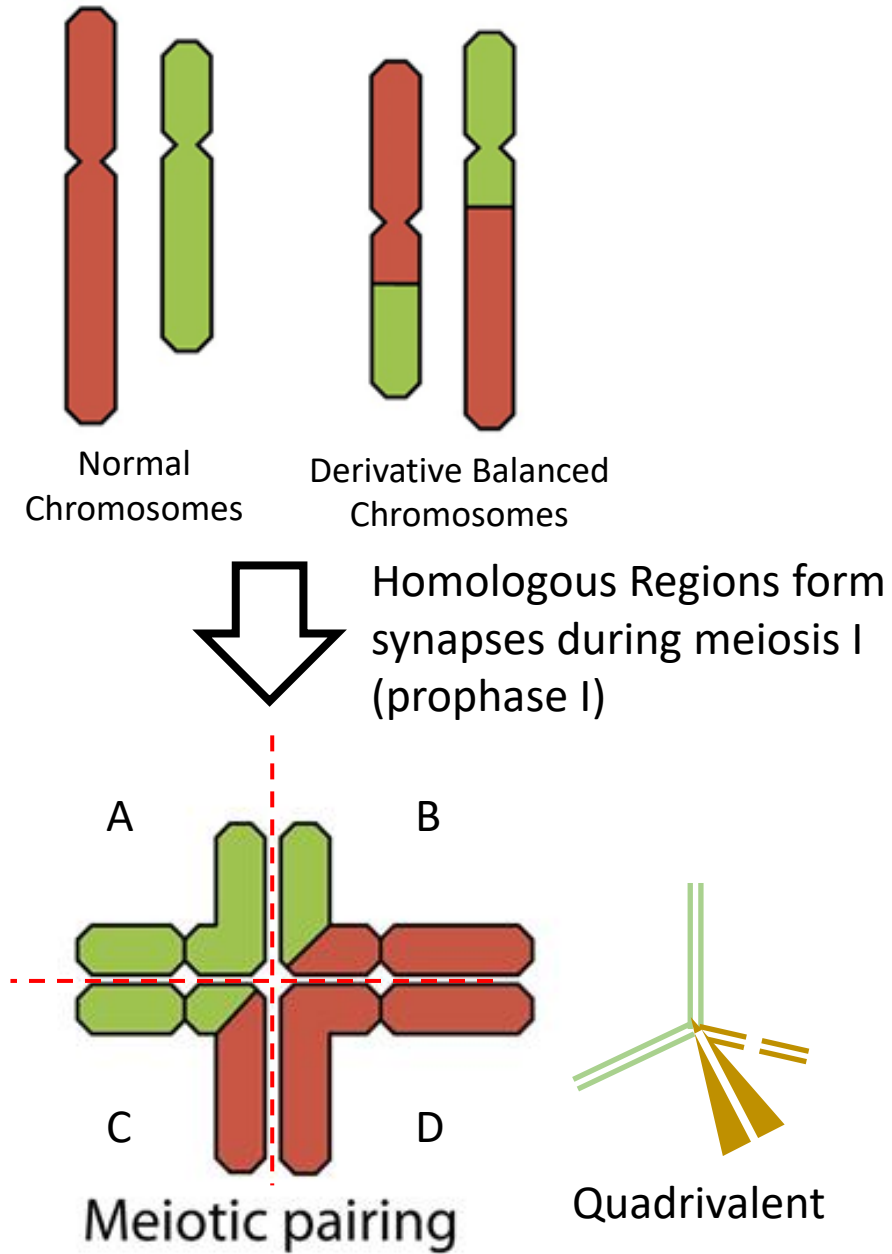
Reciprocal Translocation

What are the effects of reciprocal translocation in carriers?

- When the reciprocal translocation occurs in somatic cells, the effects may be
 - May have no effects (silent)
 - Transformation to cancer and increased growth rate of cells
- If the reciprocal translocation is present in the germ line cells, then there is a higher incidence of spontaneous abortions in the carrier



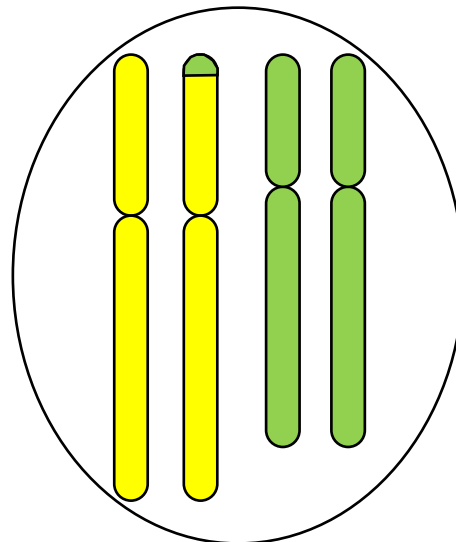
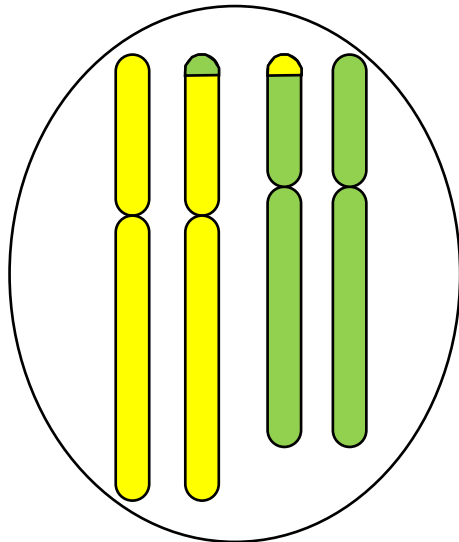
Reciprocal Translocation: Segregation at Meiosis



Case report – Reciprocal translocation

- Alice is 45-years-old with a history of repeated miscarriages and has one 21-year-old daughter named Deborah
- Deborah has just had a child named **Tracey** who has been diagnosed with Wolf-Hirschhorn syndrome
- Cytogenetic analysis of the family reveals that both Alice and Deborah are balanced reciprocal translocation carriers (Chr 4 and 8)
- Tracey has a deletion of chromosome 4 at the top of the p-arm

Alice & Deborah
Balanced genetic
material



Tracey
Partial monosomy
and partial trisomy
giving rise to Wolf
Hirschhorn

Reciprocal Translocation creating a fusion gene

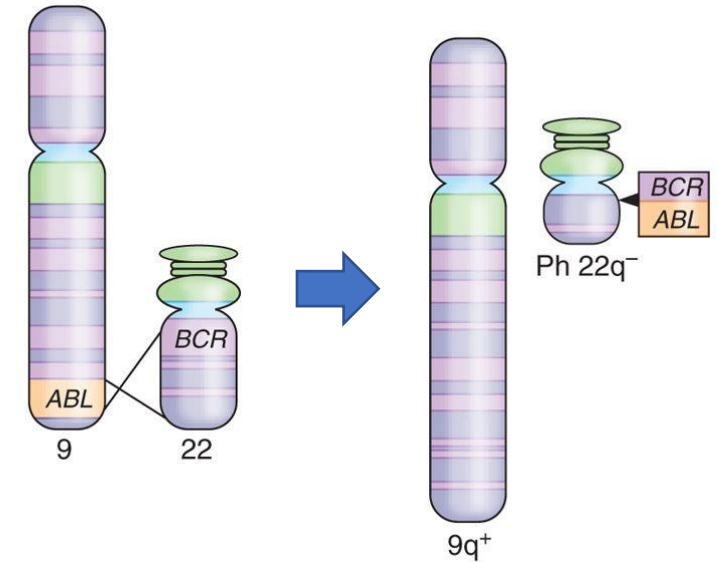
t(9;22) → Chronic myelogenous leukemia (CML)

Activation of an oncogene → Cancer

Philadelphia chromosome t(9;22) →

Activation of ***BCR-ABL*** tyrosine kinase which is a proto-oncogene in hematopoietic cells

-symptoms include fatigue, bruising, infections, hepato-splenomegaly (stimulates uncontrolled cell division)



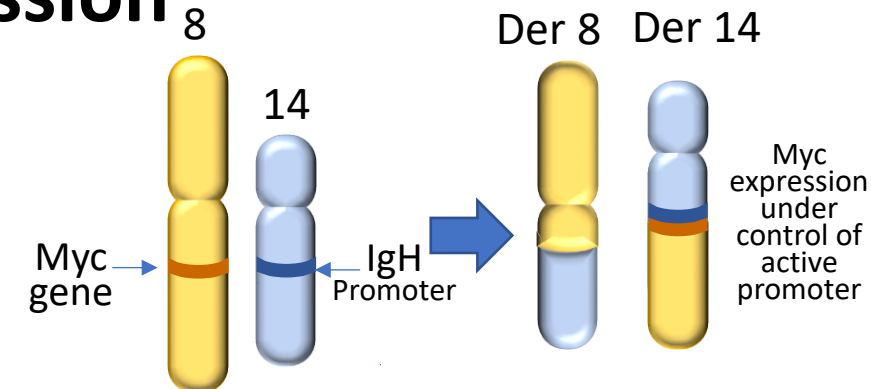
Druker, Blood 2008 112:4808-4817

Reciprocal Translocation changing gene expression

t(8;14) → Burkitt lymphoma

Dysregulation of c-Myc gene expression (↑), leading to cancer

-symptoms include enlarged lymph nodes, night sweats, weight loss (B-cells affected)

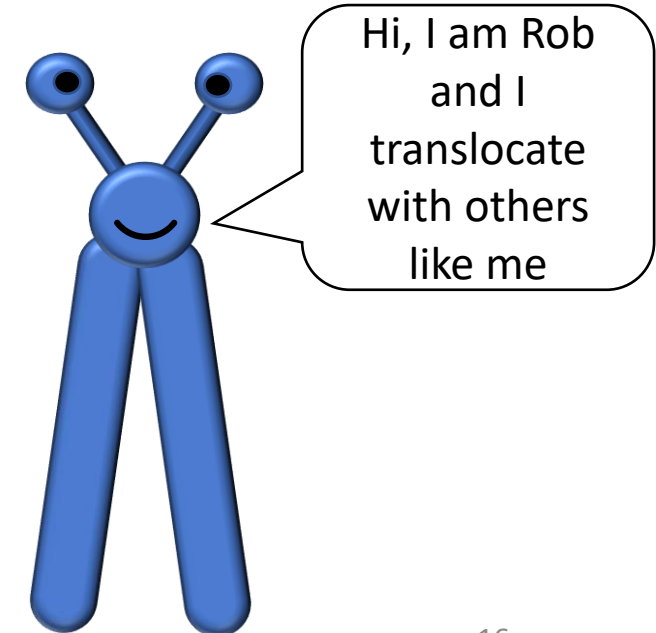
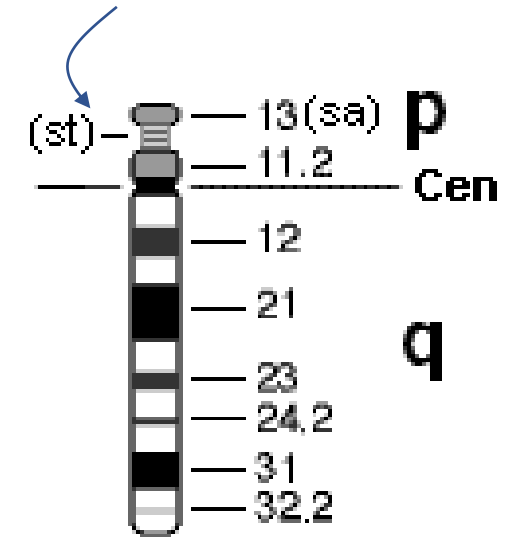


Robertsonian Translocation

- Occurs between acrocentric chromosomes
 - Chromosome 13, 14, 15, 21, 22
- There is loss of the short arms of the two chromosomes
 - The satellite structures in the acrocentric chromosomes contain multiple copies of genetic material for RNA
- Fusion of the long arms of the two chromosomes

Acrocentric chromosome

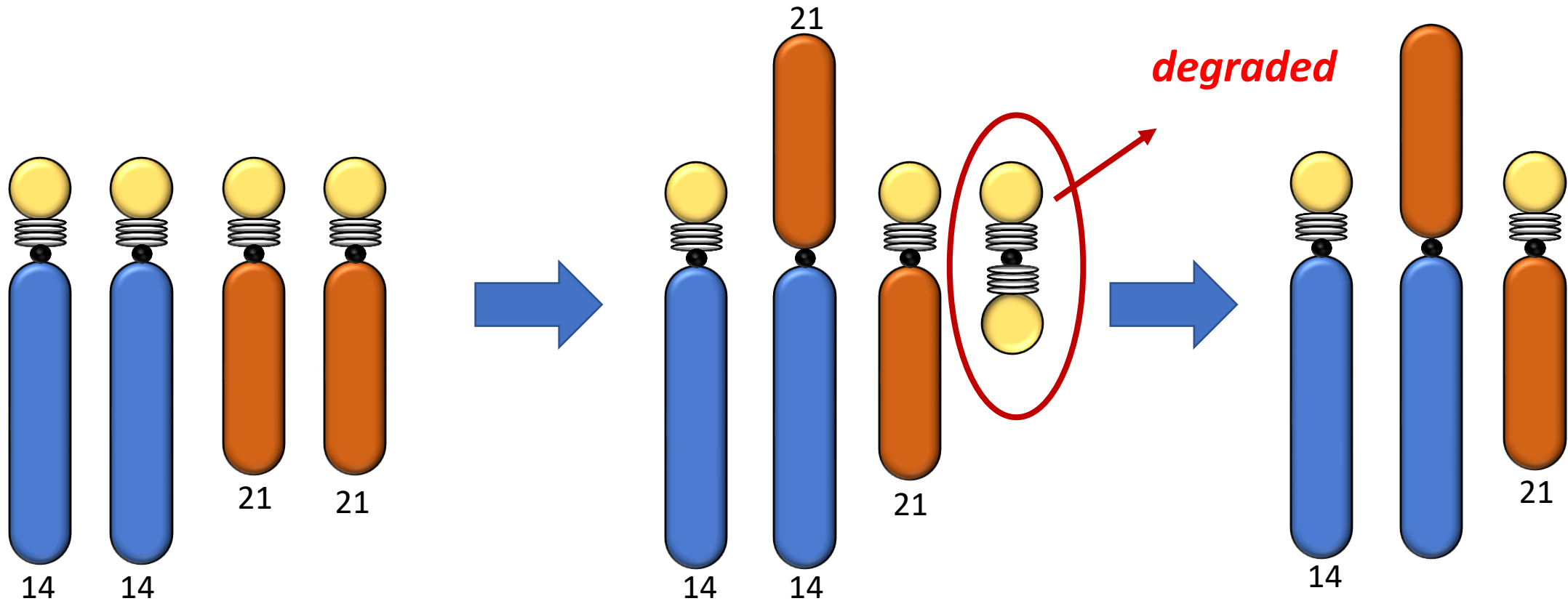
Satellite structures



Case report – Robertsonian translocation

- A 25-year-old female has a history of repeated miscarriages. She has a 2-year-old girl with Down syndrome. Karyotype analysis of the husband is normal. Karyotype analysis of the mother shows that she is a Robertsonian translocation carrier
- **45,XX, der(14;21)(q10;q10)**
- The derivative chromosome has the long arms of 14q and 21q

Robertsonian Translocation



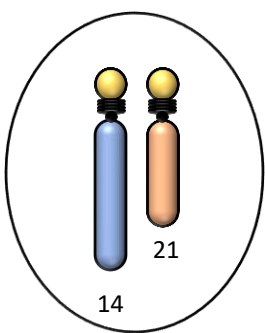
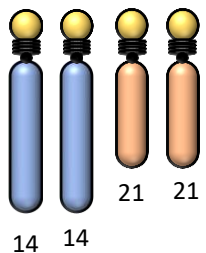
$t(14;21)$

Translocation carrier

Karyotype: 45,XX,der(14;21)(q10;q10)

Robertsonian Translocation: Familial Down Syndrome

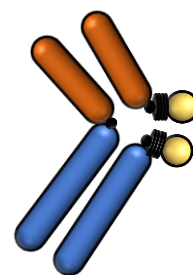
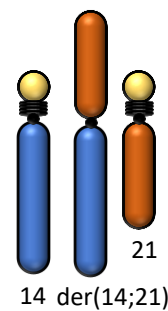
Parent with typical
chromosomal compliment
(shaded lighter for clarity)



Gamete

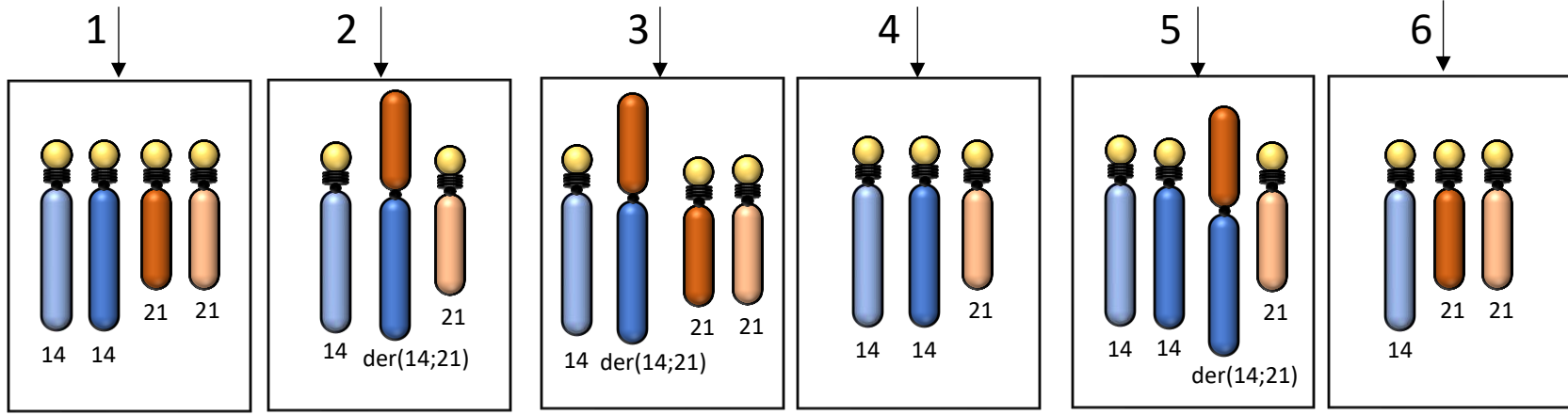
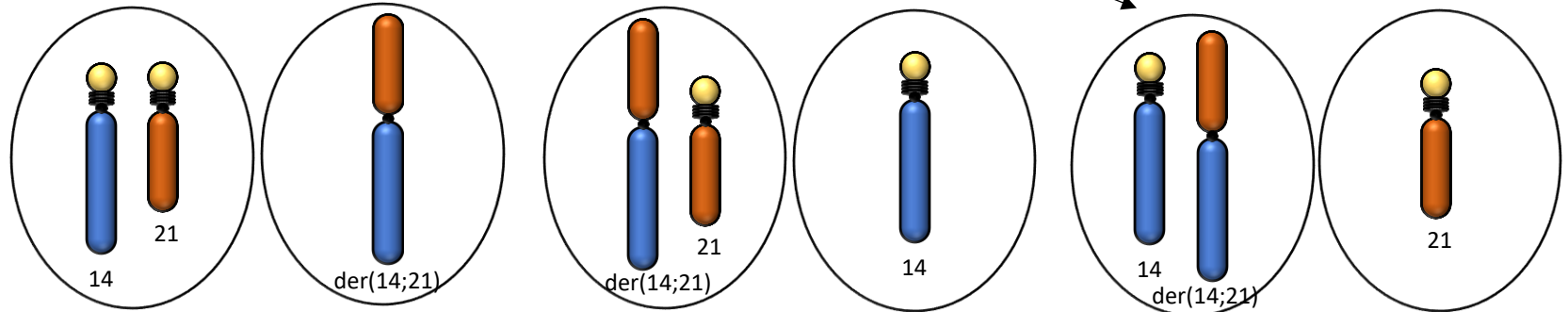
Fertilization

Carrier Parent of
Robertsonian
Translocation



Trivalent
synapsis

Possible gamete formations



Normal

Balanced
translocation

Trisomy 21

Lethal

Lethal

Lethal

Alternate Segregation

Adjacent Segregation

Robertsonian Translocation: Effects during meiosis

Alternate segregation can result in:

- Normal gamete (gamete 1)
- Gamete with the derivative chromosome (gamete 2), that results in a Robertsonian translocation carrier on fertilization

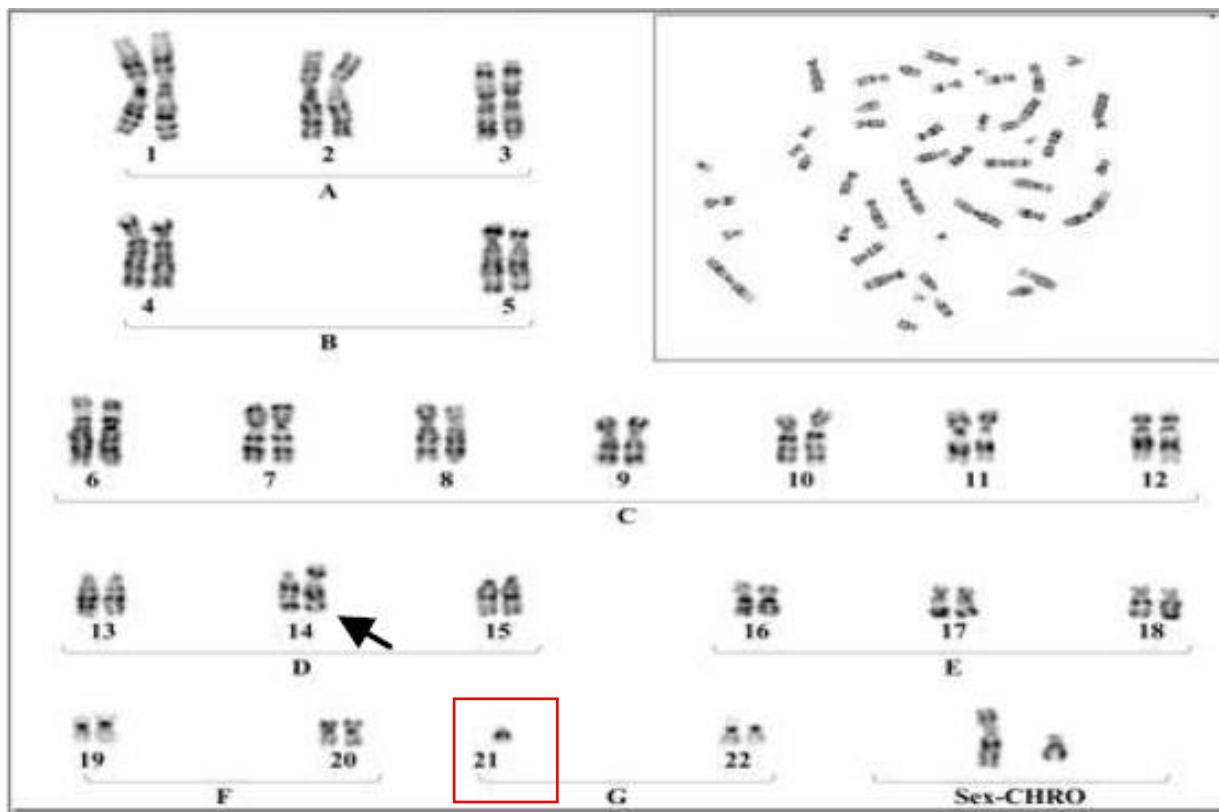
Adjacent segregation can result in:

- Gamete 3: would result in trisomy 21 (Down syndrome) on fertilization
 - Gamete 4: monosomy 21, lethal
 - Gamete 5: would result in trisomy 14, that is not compatible with survival, and results in spontaneous abortion
 - Gamete 6: monosomy 14, lethal
- Robertsonian translocation accounts for 2-5% of children with Down syndrome due to carrier parents (these carrier parents are themselves healthy)

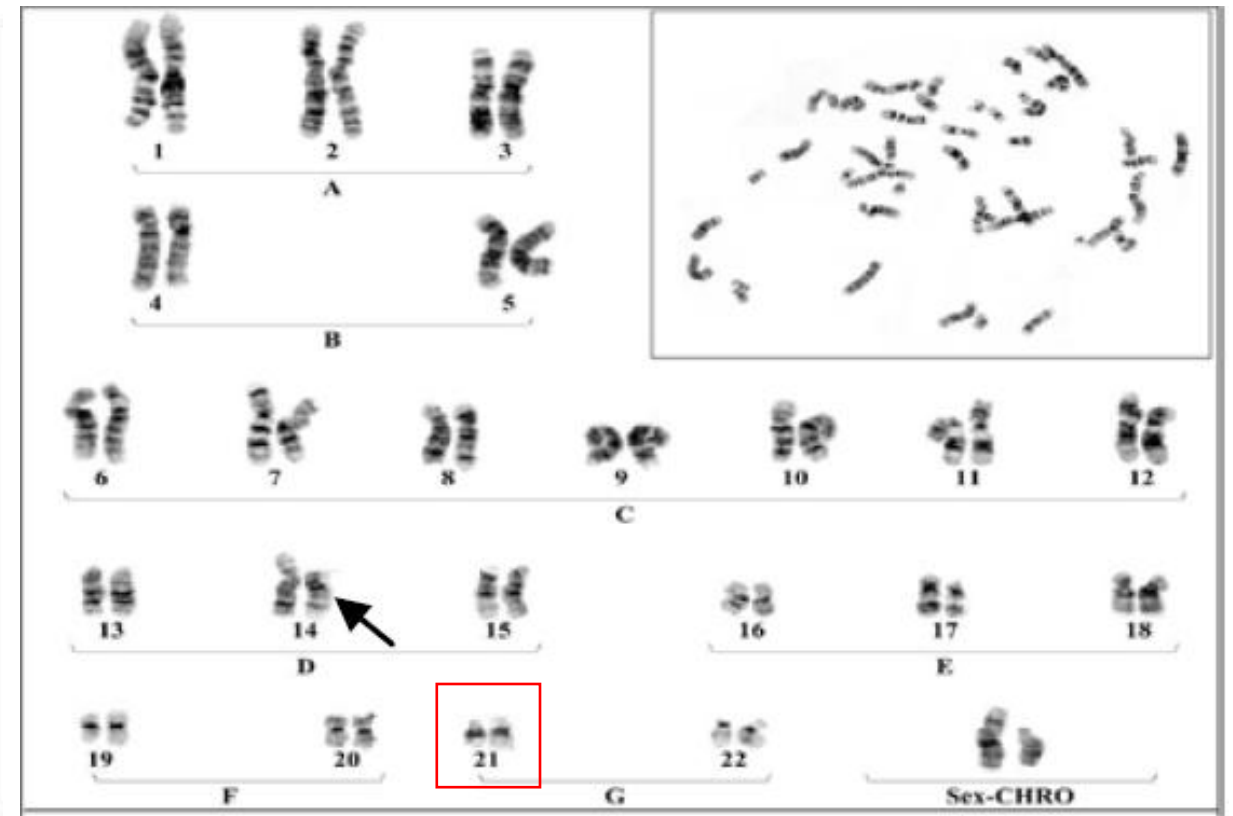
**Derivative chromosome 14 has the
long arm of chromosome 21**

**Robertsonian translocation between chromosomes
(no.21/14) in relation to the history of spontaneous
abortion in a family**

Mohammad Hasanzadeh-NazarAbadi Ph.D., Fatemeh Baghbani M.Sc., Iman Namazi M.D., Salmeh Mirzaee M.Sc.



Carrier 45,XY,rob(14:21)(q21.1;q11.2)



Proband 46,XY,rob(14:21)(q21.1;q11.2)

Indications for Cytogenetic or Cytogenomic Testing for a microdeletion syndrome:

- Unexplained developmental delay or autism spectrum disorder
- Congenital anomalies/dysmorphic features
- Suspected deletion or duplication syndrome



Pathognomonic/ Dysmorphic features
Are influenced by
Genetic/Ethnic background

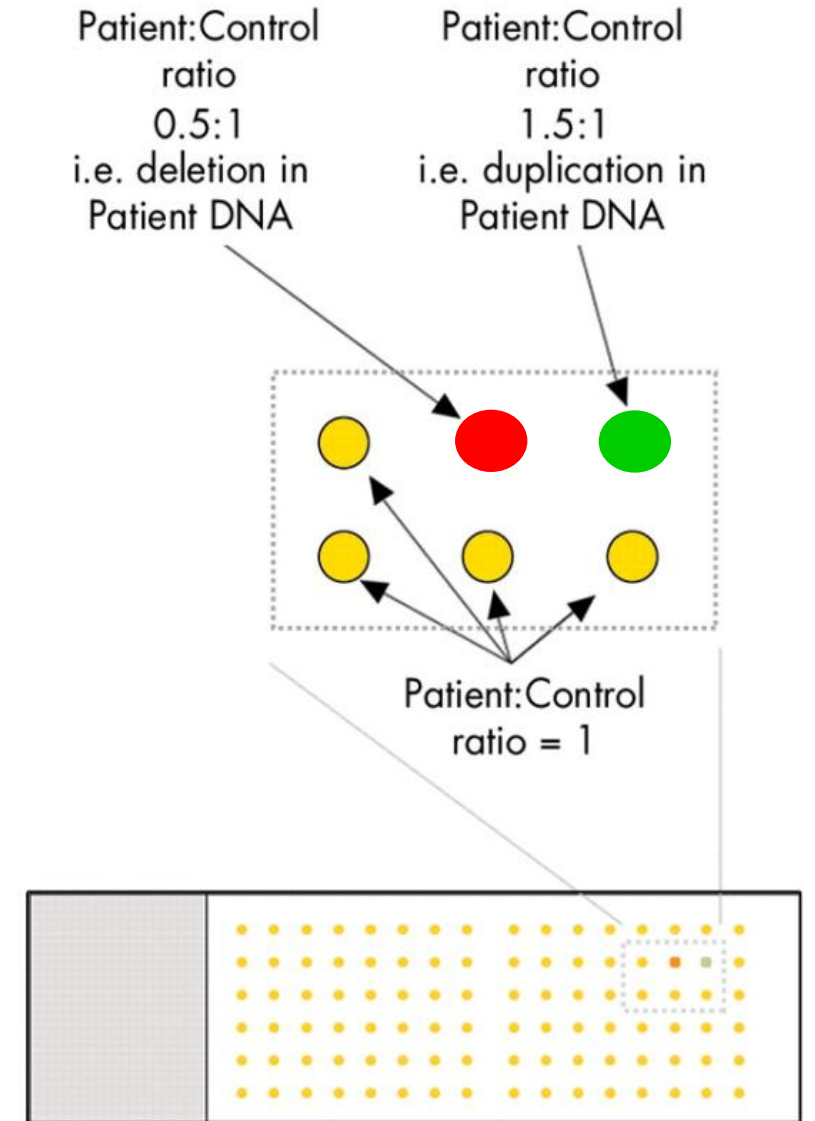
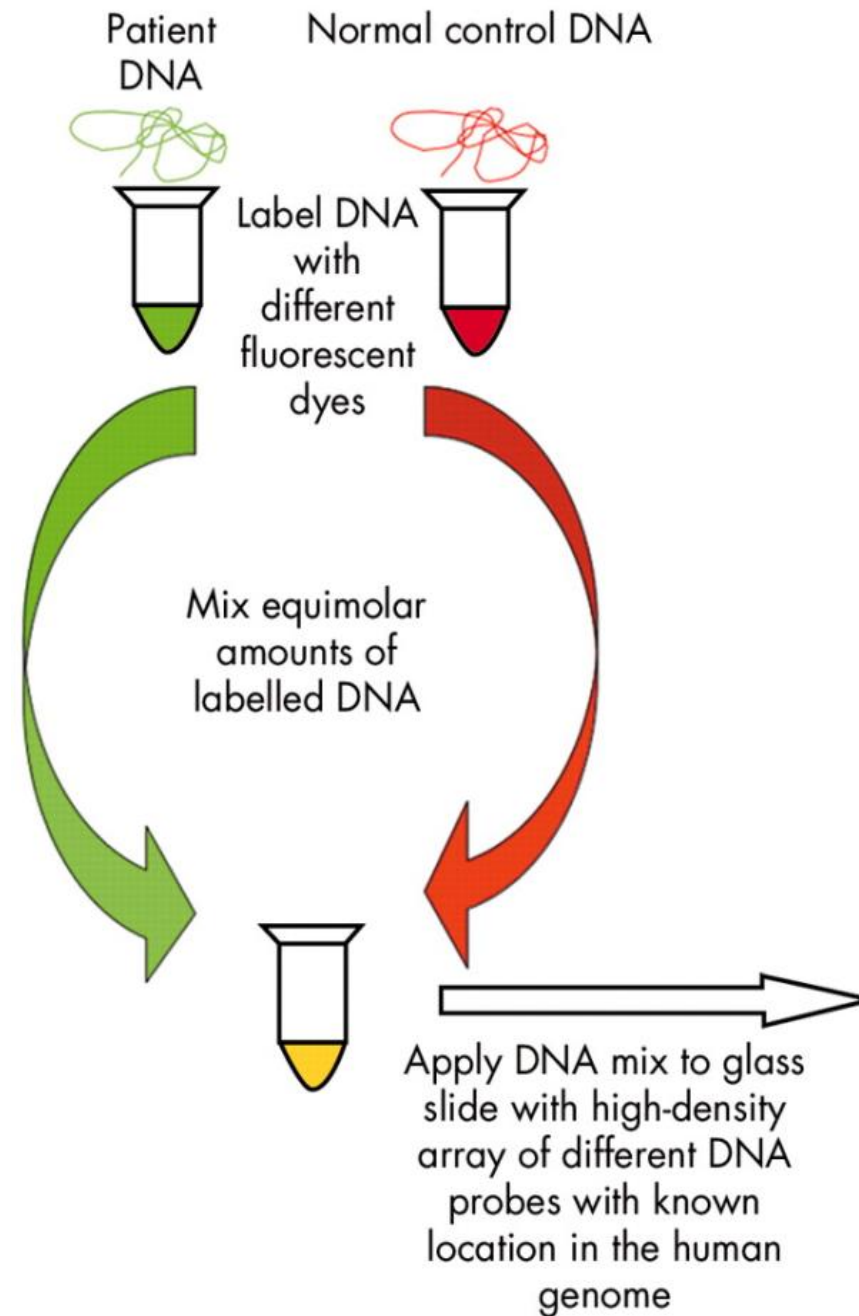
Deletion 17q12 Is a Recurrent Copy Number Variant that Confers High Risk of Autism and Schizophrenia
Moreno-de-Luca *et al.* 2010, The American Journal of Human Genetics
Macrocephaly, epicanthal folds, downslanting palpebral fissures, arched and high eyebrows, slightly depressed nasal bridge, and malar flattening are the features shared most consistently among these patients.

Microdeletion Syndromes

- Large deletions
 - Approximately 5 Mb or larger are usually detected by standard karyotype
 - Must affect banding pattern though
 - Some very large deletions (~10Mb) appear normal by karyotype
 - Some very large duplications may appear normal by karyotype
- Small deletions
 - Smaller than 4 Mb may not be detected by standard karyotype, called “Microdeletions”
- Well characterized common deletion syndromes have historically been tested by FISH
- Microarray CGH can now be used to detect common deletion syndromes, and can detect uncharacterized very small deletions and duplications

Microarray CGH

- Each spot on the array is an immobilized probe for a specific place in the genome
- Allows the use of almost a million probes (high density)



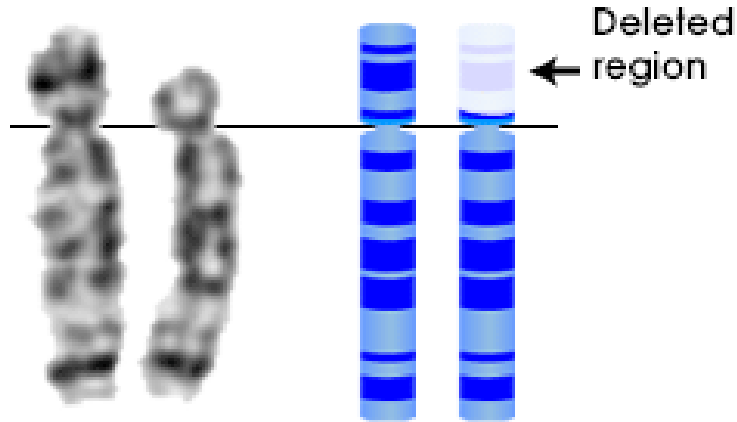
Cri-du-chat Syndrome

- Deletion of chromosome 5p
 - **Cri-du-chat** syndrome
- 46,XX,del(5)(p15.3)(pter) or 46,XY,del(5)(p15.3)(pter)
- Features:
 - High pitched, cat-like cry
 - Micrognathia which influences the cat-like cry and later, speech problems
 - Severe intellectual disability
 - Microcephaly
 - Hypertelorism (widely spaced eyes)



Cri-du-chat syndrome diagnostic tests

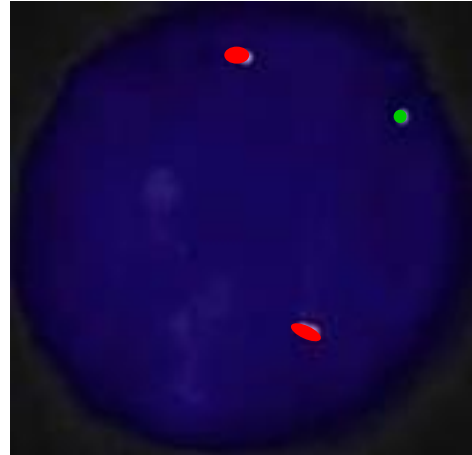
CGH for diagnosis of cri du chat syndrome: Array CGH revealed a 3.6Mb deletion at 5p15.33-13.3



Cri-du-chat Chromosome 5 pair

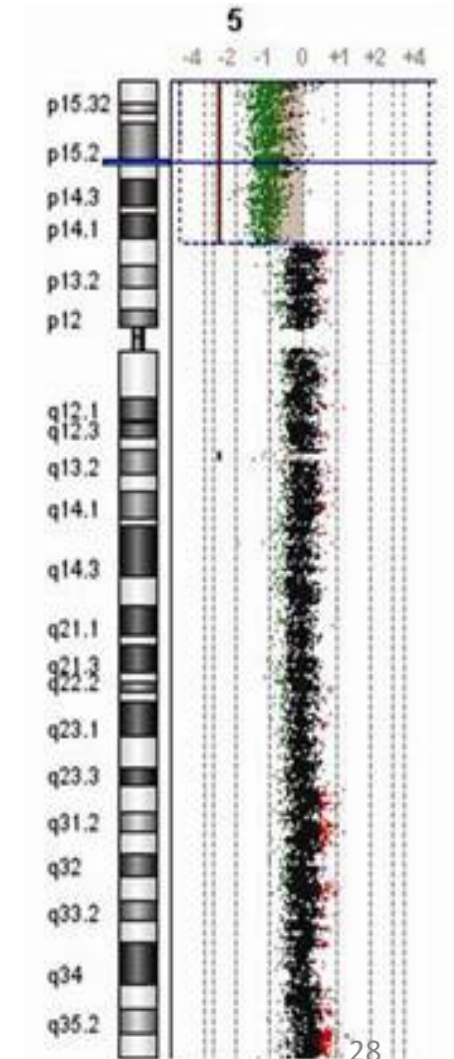
Sometimes the deletion that leads to cri du chat syndrome will be detected by karyotype, but not always.

FISH for diagnosis of cri-du-chat syndrome



Med & Health 2010; 5(2): 108-113

Interphase FISH analysis shows one (1) green and two (2) orange signal pattern, indicating the deletion of one 5p15.2 region.



22q11.2 Deletion Syndrome aka: Velocardiofacial syndrome or DiGeorge

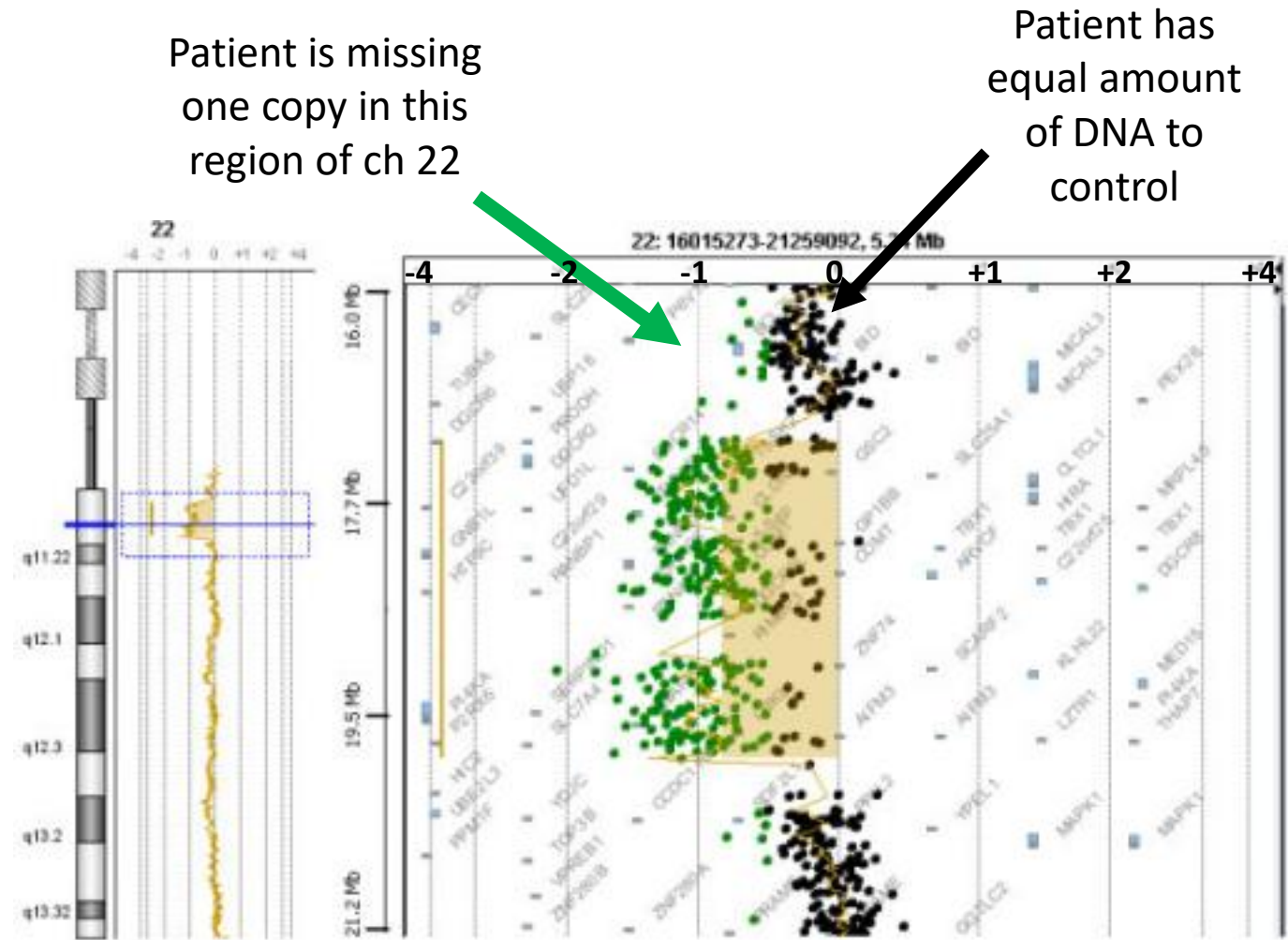
Most common contiguous gene (microdeletion) disorder in humans, ~1:4,000 to 1:6,000



- **Microdeletion of chromosome 22q11.2**

- Congenital heart defects
- Absence of thymus (thymic aplasia) – might cause Immunological problems
- Cleft lip & palate; maybe speech difficulty
- Learning disability
- Facial anomalies include long midface, narrow palpebral fissures, prominent nasal root and bulbus nasal tip, ear dysmorphology
- Increased risk for schizophrenia
- Genes in deleted region interact with Drosha in miRNA regulation

aka Velocardiofacial syndrome or DiGeorge syndrome



Wolf-Hirschhorn syndrome (WHS)

- Deletion of ch 4p
- Seizures
- Skeletal abnormalities and congenital heart defects
- Spectrum of intellectual and developmental delay, mild to severe
- Facial anomalies - widely spaced eyes, high-arched eyebrows, broad & flat nasal bridge often described as similar to a roman helmet, short philtrum, downturned mouth, small chin



Image courtesy of John Carey, University of Utah;
used with permission

<http://medgen.genetics.utah.edu/photographs/pages/wolfhirschhorn.htm>



Wolf-Hirschhorn syndrome



- Think before you speak:
 - Survey of parents who have WHS children reveals that they feel uncomfortable, hurt, offended when their physician describes their baby as having a roman helmet shaped forehead

Image courtesy of John Carey, University of Utah; used with permission

Deletions in Chromosome 15q11-13

- A few genes on Chromosome 15 are monogenic (only one allele is expressed)
- Two genes in PARTICULAR are SNRPN and UBE3A
- One copy of each of these genes are methylated, thus silenced
- Prader Willi syndrome & Angelman syndrome are due to deletions of chromosome 15q11 to 15q13

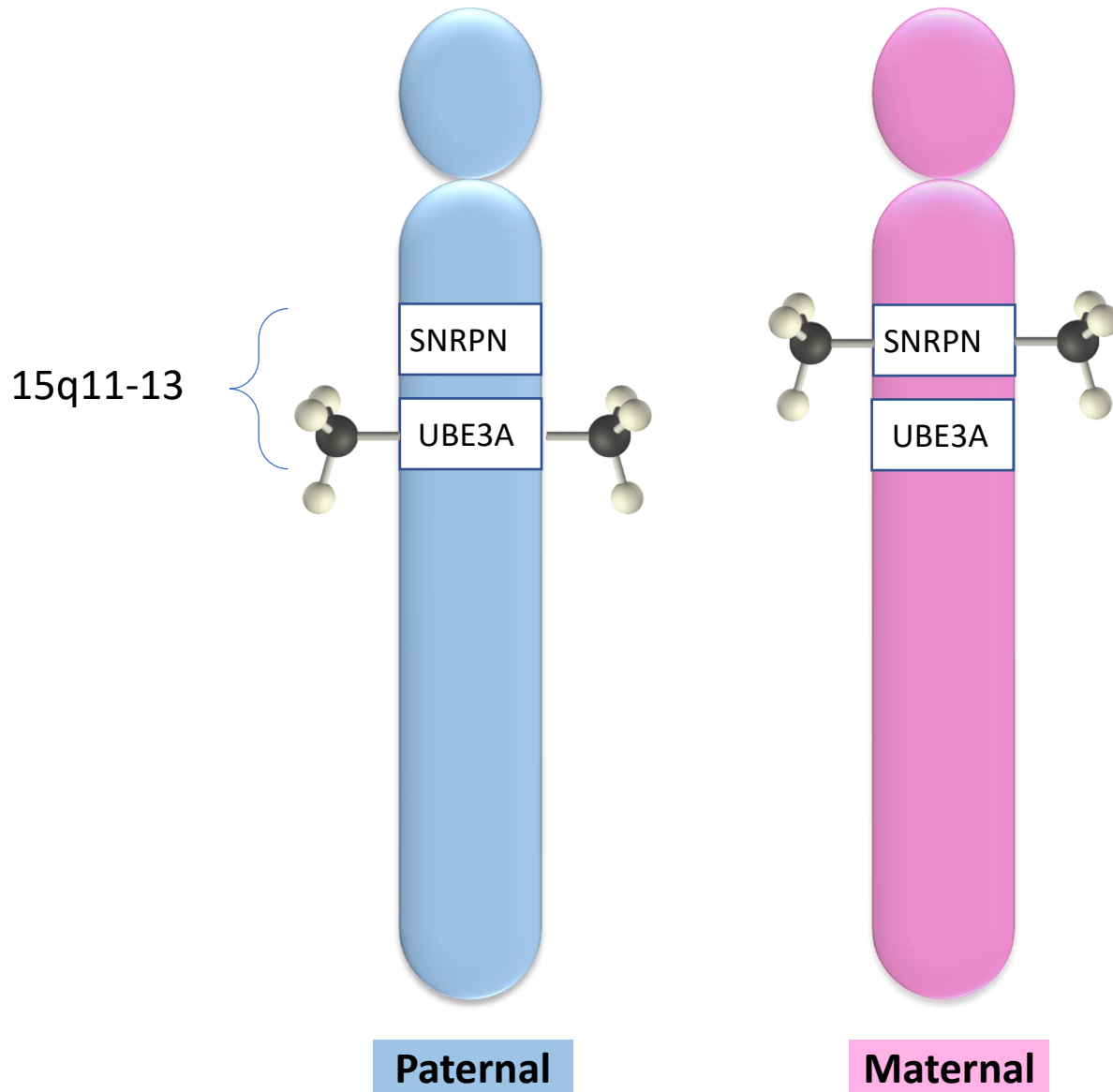
- **P**rader-Willi syndrome is due to deletion of **P**aternal chromosome 15q11-13 which includes SNRPN gene

Maternal gene SNRPN is methylated, thus “turned off”

- Angel**M**an syndrome is due to deletion of **M**aternally chromosome 15q11-13 which includes the UBE3A gene

Paternal gene is UBE3A methylated, thus “turned off”

Chromosome 15q11.1-13 under *Normal Conditions*



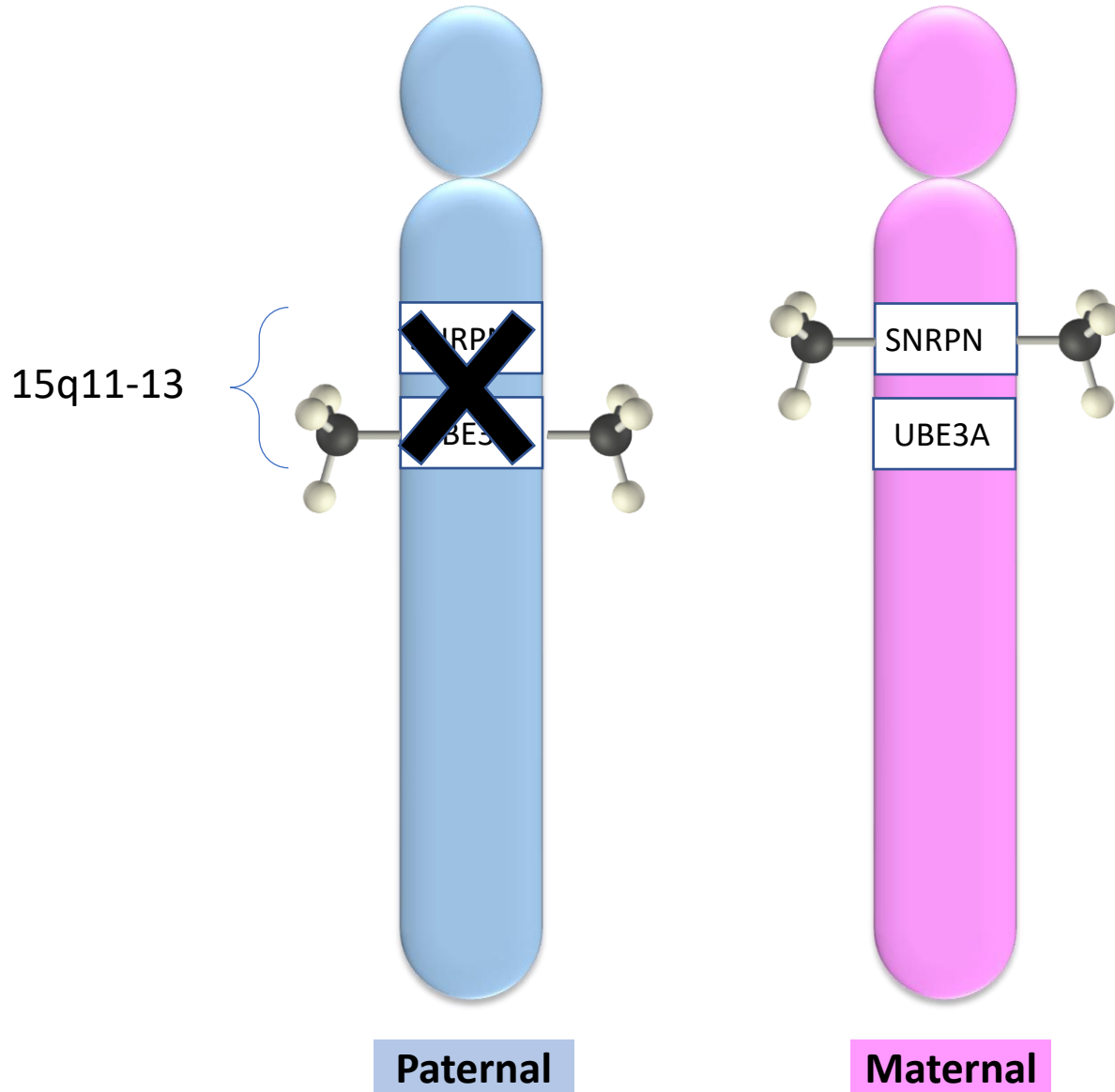
SNRPN GENE

One copy, the paternal copy of the SNRPN gene is expressed in a person (maternal copy is methylated and turned off)

UBE3A GENE

One copy, the maternal copy of the UBE3A gene is expressed in a person (paternal copy is methylated and turned off)

Chromosome 15q11.1-13 Deletion Syndromes



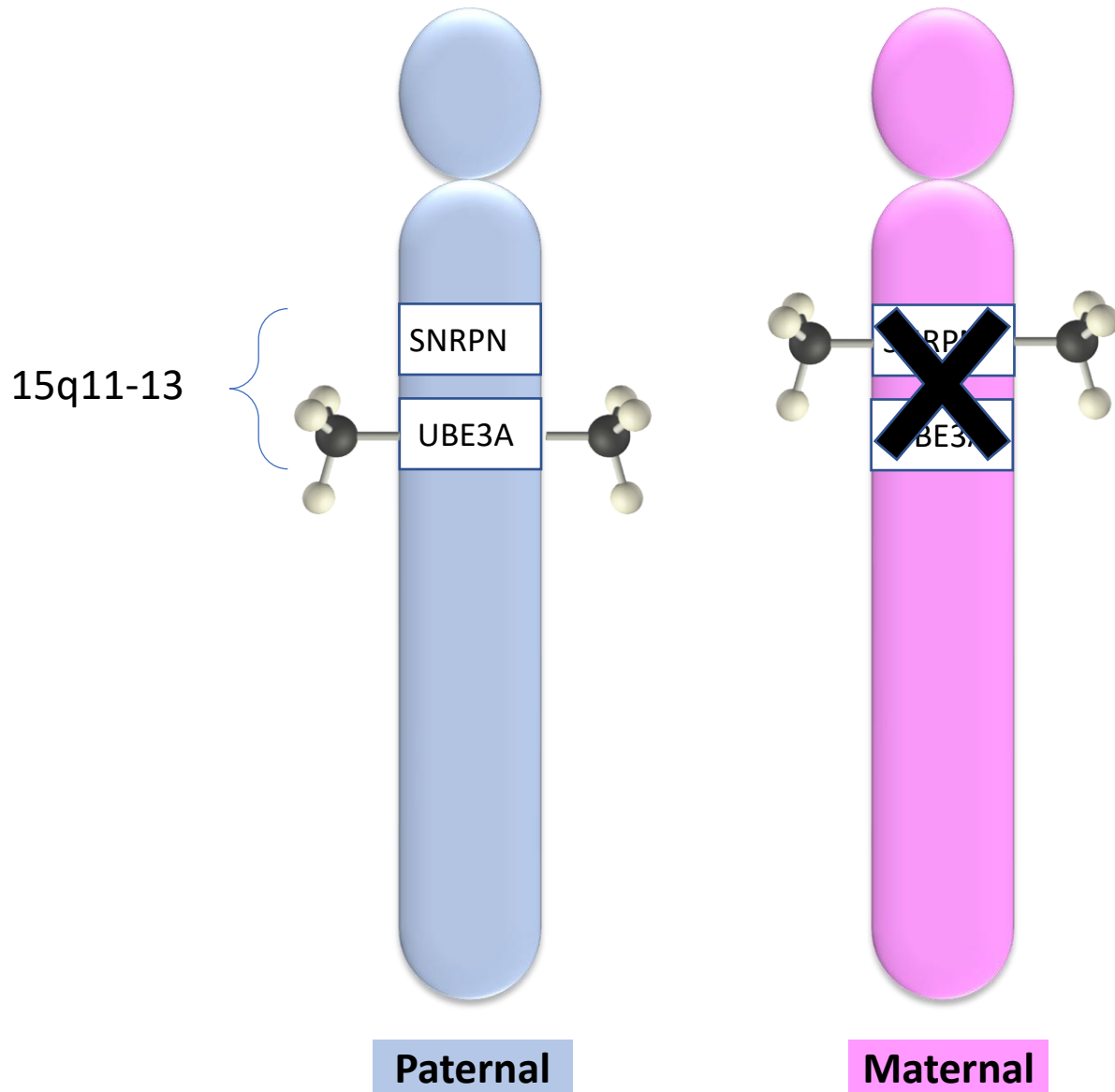
SNRPN GENE

The **P**aternal copy of the **SNRPN** gene is **D**ELETED in a person (maternal copy is methylated)

= **P**rader Willi Syndrome

Mnemonic: PPDP

Chromosome 15q11.1-13 Deletion Syndromes

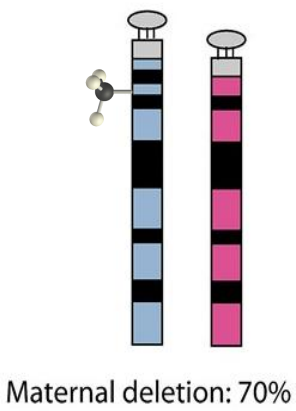
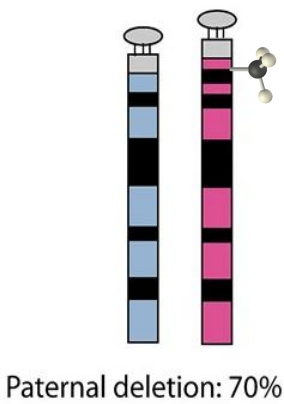
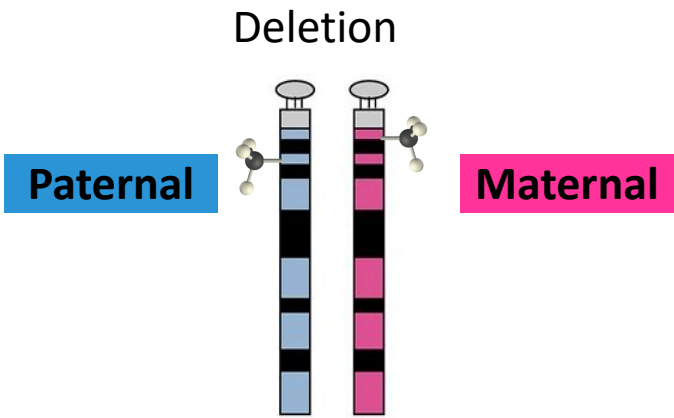


UBE3A GENE

The **IV** maternal copy of the **UBE3A** gene is **D**ELETED in a person (paternal copy is methylated)

= **A**ngel**IV**an Syndrome

Mnemonic: MADAM

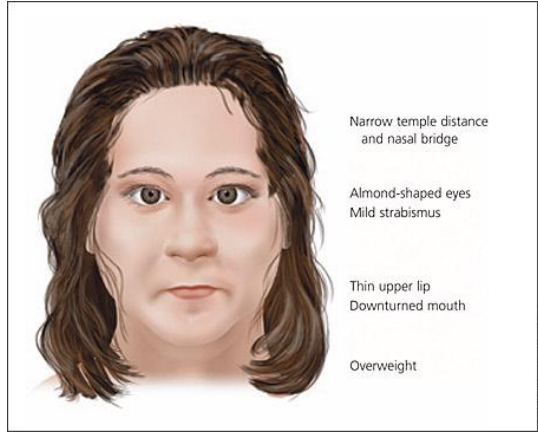


Normal Chromosomes 15

shown with G-banding pattern

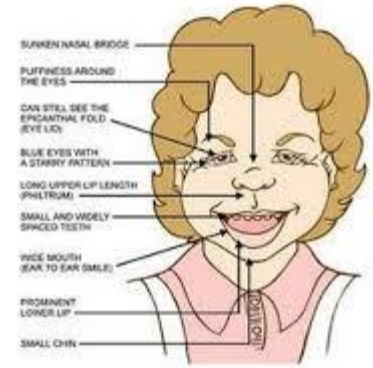
Prader Willis Syndrome

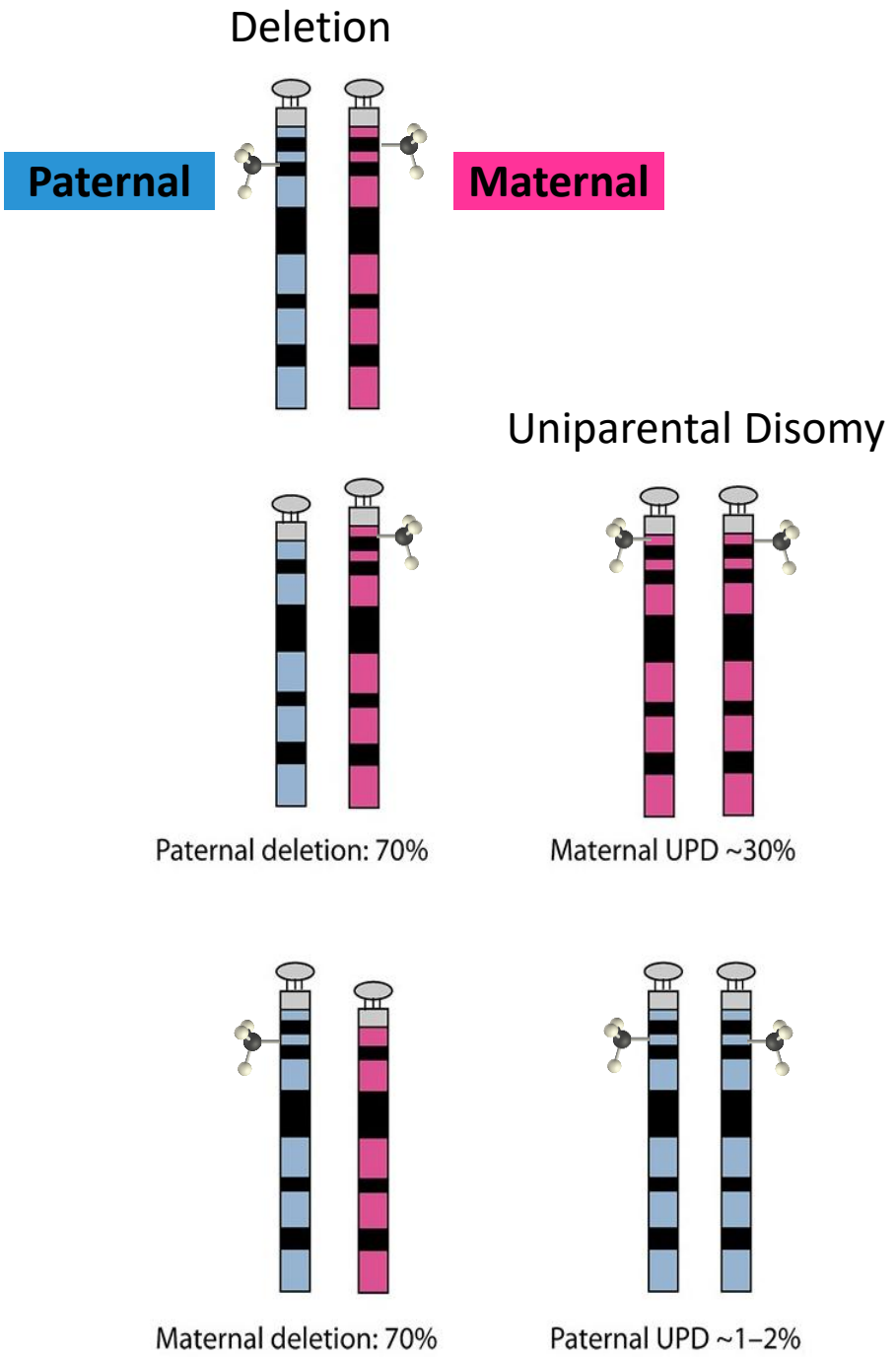
- Eating disorder- hyperphagia
- Developmental delay
- Hypotonia at birth
- Hypogonadism
- Distinctive facial features



Angelman Syndrome

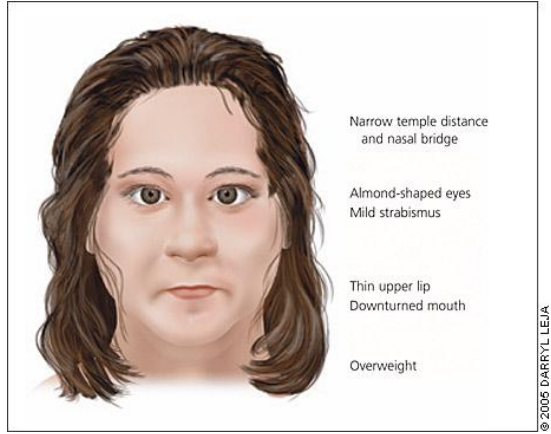
- Unprovoked laughing and smiling
- Jerky uncoordinated movements
- Lack of speech
- Hypotonia
- Severe developmental delay



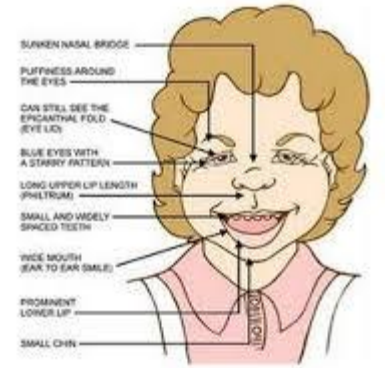


Normal Chromosomes 15
shown with G-banding pattern

Prader Willis Syndrome
Eating disorder- hyperphagia
Developmental delay
Hypotonia at birth
Hypogonadism
Distinctive facial features



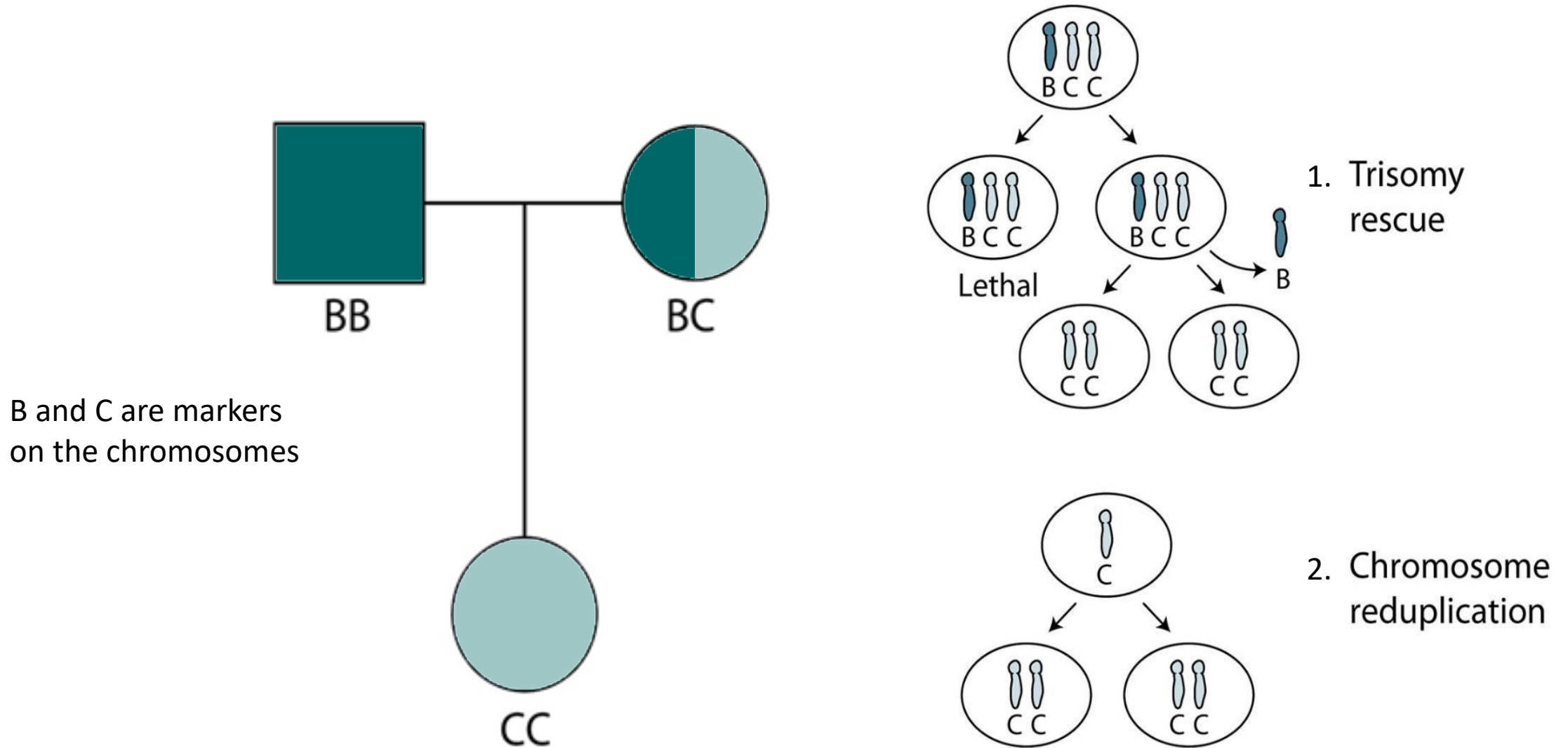
Angelman Syndrome
Unprovoked laughing and smiling
Widely spaced teeth
Jerky uncoordinated movements
Lack of speech
Hypotonia
Severe developmental delay



Uniparental Disomy of Chromosome 15

- Some cases of Prader Willi and Angelman syndrome are due to uniparental disomy (inheritance of both chromosomes 15 from same parent)
 - Uniparental disomy might occur by **trisomy rescue mechanism**
 - Non-dysjunction of chromosomes while developing gamete, then upon fertilization there is a trisomy. The zygote may try to correct by kicking out one the excess chromosomes
 - Uniparental disomy might occur if a fertilized embryo is monosomic. The single chromosome that remains might be duplicated to recreate the diploid state.

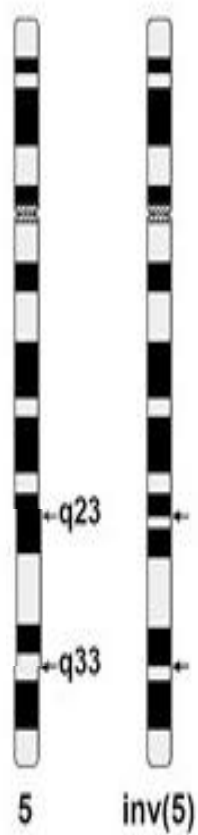
Uniparental Disomy: Two Theories



In this case, uniparental disomy is shown as occurring from the mother because the markers are maternally derived

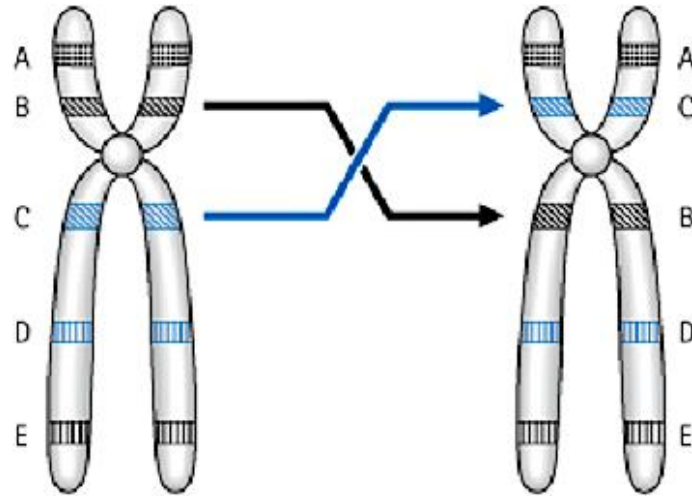
A few more structural abnormalities:

Inversions

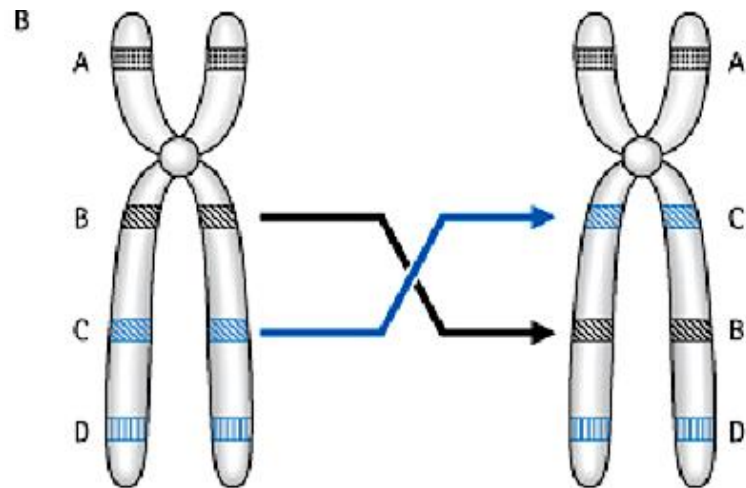


- May be **p**ericentric (involves the **c**entromere) or paracentric (does not involve the centromere)
- Usually balanced & no clinical problems in carriers
- Inversions usually result in a change in the banding pattern of the chromosome and can be identified by karyotype analysis

Pericentric or Paracentric Inversions

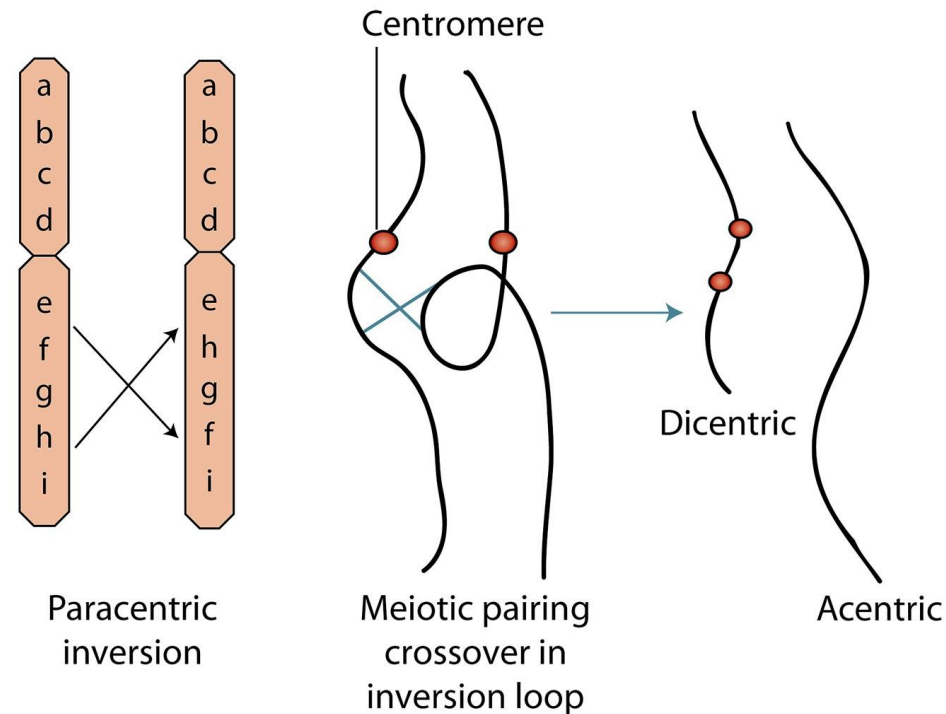


Pericentric inversion



Paracentric inversion

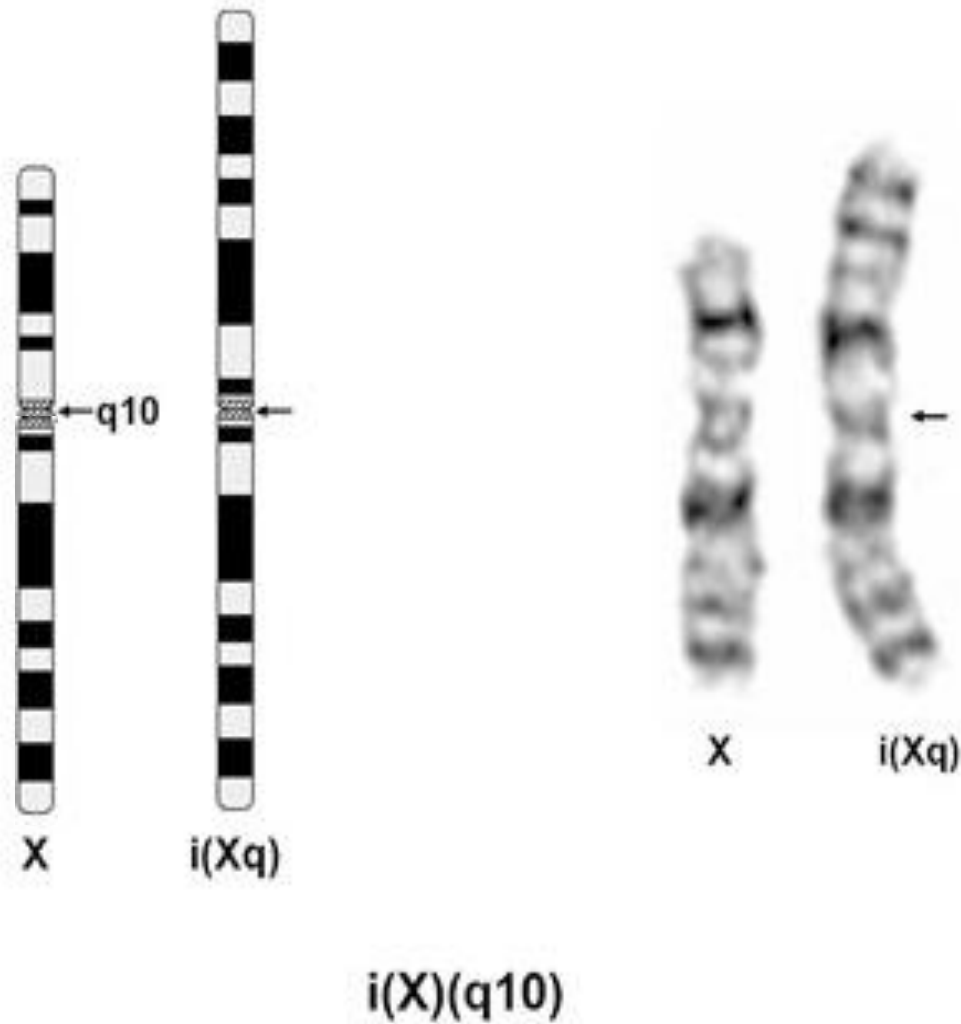
Other Effect of Inversions During Crossover in Meiosis



Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.
© 2013 John Wiley & Sons, Ltd. Published 2013 by John Wiley & Sons, Ltd.

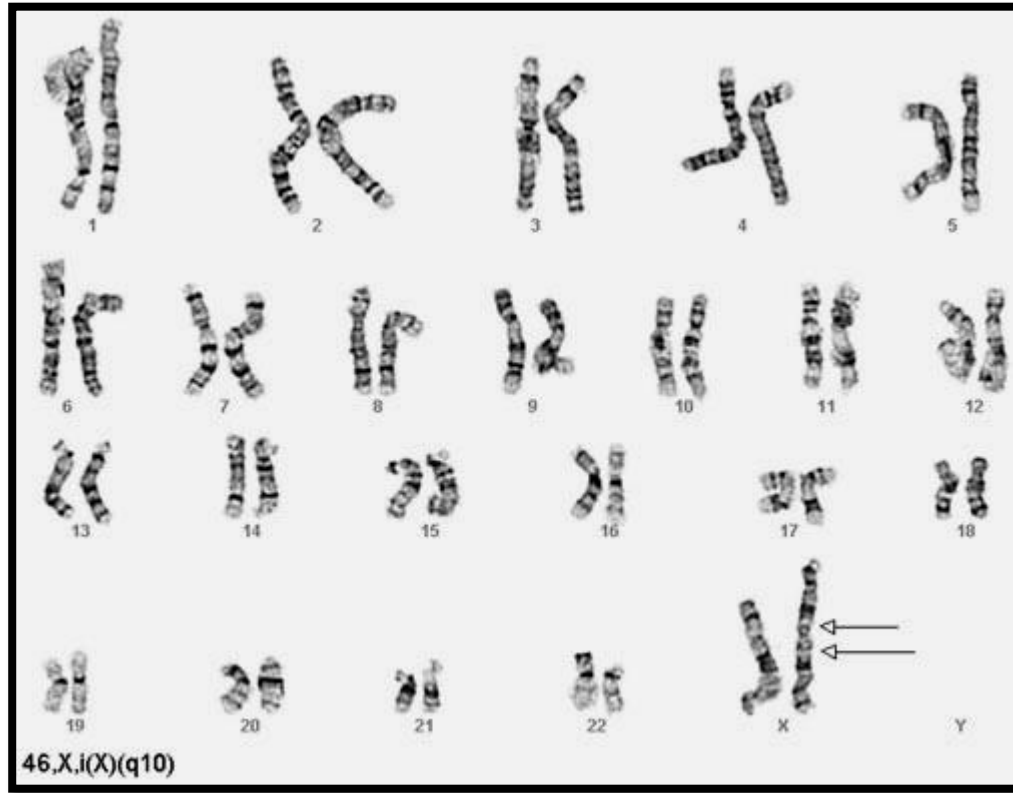
Crossover during meiosis results in duplication or deletion of segments of the chromosome (unbalanced genetic material). The gametes with acentric and dicentric chromosomes are not viable. Carriers of inversions may be asymptomatic, but they have a high risk of **spontaneous abortions**.

Isochromosomes



- There is loss of one arm of a chromosome & duplication of the other arm
- X isochromosome: long arms of the X chromosome join to form an isochromosome
 - Girls with isochromosome X have Turner syndrome
- Typically results in chromosomal and gene dosage imbalance
- Isochromosome of an autosome is lethal
 - May be tolerated if mosaic

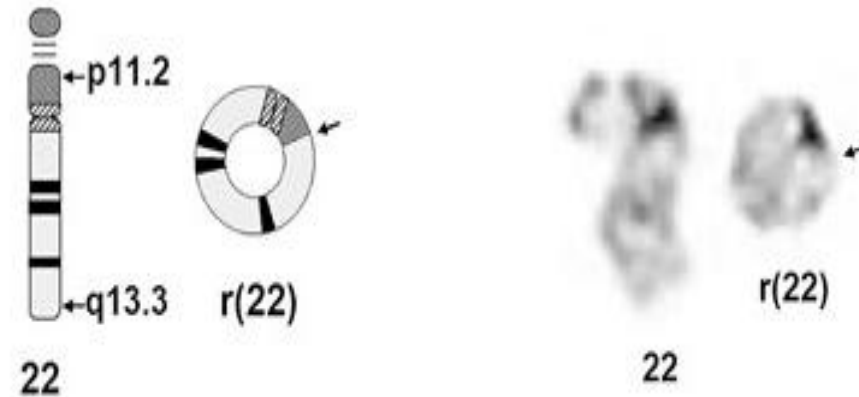
X-isochromosome i(X)



- Some children with Turner syndrome have an isochromosome of one of the X chromosomes
- They have only one copy of the short arm of X chromosome
- The isochromosome is typically inactivated,
- Haploinsufficiency of genes on the p arm that escape X inactivation thought to give rise to phenotype

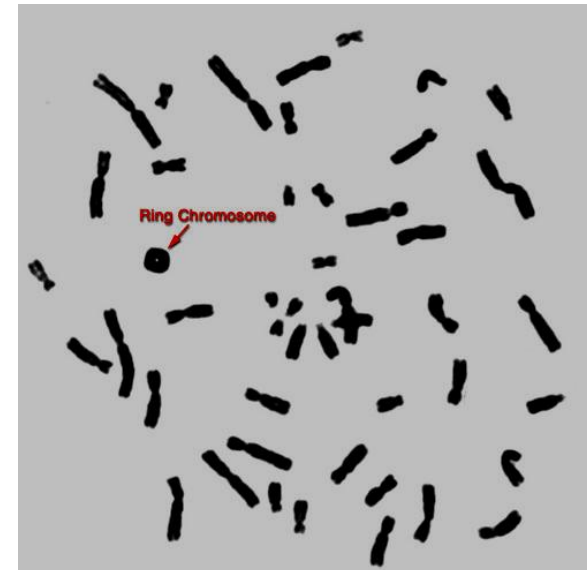
Ring Chromosomes

A ring chromosome forms when a chromosome loses genetic material at the terminal portions & the ends fuse to form a ring like structure



r(22)(p11.2q13.3)

In the radiation worker for whom this cell culture was evaluated for structural chromosome aberrations, a total of 4 cells each containing one ring were identified in a sample of only 100 cells.



Some specific points for the student to think about while studying

- Could classify chromosomes based on morphology and the basis of ISCN nomenclature
- Could identify the laboratory techniques used for diagnosing cytogenetic disorders such as: karyotyping, CGH, and FISH
- Could outline the following types of balanced and unbalanced structural rearrangements:
 - Compare / contrast reciprocal and Robertsonian translocations
 - Define and then distinguish between deletions, inversions, ring chromosomes and isochromosomes
 - Indicate the difference between pericentric and paracentric inversion
 - describe how balanced rearrangements can give rise to unbalanced rearrangements in the next generation (due to meiotic segregation)
- Might think about identifying the most common chromosomal abnormality present in chronic myeloid leukemia and in Burkitt lymphoma.
 - Relate the effect of the translocation to tumorigenesis and revisit after lectures in Cancer Genetics
- Should identify possible gametogenesis effects in a translocation carrier
- Could give some examples of microdeletion syndromes and describe their genetic basis and clinical presentations - Cri-du-chat syndrome, 22q11.2 del syndrome), and Wolf-Hirschhorn syndrome
- Think about the clinical features, the typical karyotype, and the common name associated with the aneuploid conditions described above
 - Trisomy 21, Trisomy 13, Trisomy 18
 - Triploidy, tetraploidy
 - Turner syndrome, Klinefelter syndrome
 - Identify the genetic mechanism responsible for the aneuploidy

Some more specific points for the student consider while studying

- Predict the relationship between chromosome abnormalities and pregnancy loss
- Consider the relationship between maternal age and fetal aneuploidy
- Distinguish, analyze, compare and contrast the types of chromosomal abnormalities that can give rise to Down syndrome and the implications of each for predicting recurrence risk – include a discussion of:
 - Aneuploidy due to non-disjunction
 - Aneuploidy due to Robertsonian translocation
 - Down syndrome as a result of mosaicism
- Give some examples for the clinical applications of single locus Fluorescent *in situ* Hybridization (FISH)
- Differentiate between the effects of nondisjunction during gametogenesis and nondisjunction that occurs during embryogenesis
- Explain the genetic basis of mosaicism, and recognize how trisomy rescue can lead to mosaicism in the fetus
- Predict whether a nondisjunction event occurred during meiosis I or meiosis II based on genetic analysis

The End

